

Reasoning or Remembering? A Contamination Aware Reliability Evaluation of a Fully Open Language Model Family

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ABSTRACT

Large language models achieve strong scores on reasoning benchmarks, yet accuracy alone cannot reveal whether a model performs genuine inference or retrieves memorised solutions. To our knowledge this is one of the first unified evaluations to combine direct contamination detection against open training data, consistency based reliability scoring across semantically equivalent variants, and lightweight training data attribution on a single fully open reasoning specialised model family. We focus on the AMD Instella family because its weights, training mixtures, and post training corpora are publicly documented. Using four reasoning benchmarks and two perturbation types, we compute a Reliability Score defined as the product of cluster accuracy and the agreement of correctness labels across variants, and pair it with embedding nearest neighbour attribution against the documented training data. The result is a small but defensible diagnostic profile that separates likely memorisation from likely generalisation per reasoning subskill, fully runnable on a single consumer grade GPU.

PROBLEM STATEMENT

Modern reasoning evaluation is dominated by accuracy on fixed benchmarks. Perturbation studies have shown that headline performance can collapse when an irrelevant clause is added to a standard arithmetic problem [2], indicating that a substantial share of accuracy may reflect pattern matching rather than reasoning. The debate is unresolved because the evidence remains indirect. Most frontier models keep their training data proprietary, so researchers cannot test whether benchmark items, or close paraphrases of them, were present during pretraining. Existing detection approaches rely on perplexity heuristics or surface n gram overlap, both of which are noisy and miss paraphrased contamination.

A second limitation is that three relevant research threads remain largely separate. Reasoning benchmarks measure accuracy and, increasingly, robustness, but cannot distinguish capability from contamination without training data access [1, 2]. Memorisation studies quantify how often training items are reproduced, but do not connect this directly to reasoning quality [3]. Training data attribution traces outputs back to influential training examples [4], yet is rarely applied to reasoning capability on a fully open reasoning specialised model. We address this gap on a focused subset of benchmarks within a clearly bounded compute budget, treating reasoning versus remembering as a testable hypothesis rather than asserting a settled finding.

PROPOSED METHODOLOGY

We propose a four phase pipeline designed to fit on a single consumer GPU. The subject of study is the AMD Instella family, which provides multiple model sizes trained on the same documented data pipeline.

Model	Parameters	Role in Study
AMD OLMo 1B	1.2 B	Smaller open companion; used for the gradient attribution validation step in full precision
Instella 3B	3 B	Base reasoning model; primary subject of the contamination and reliability analysis
Instella 3B Math	3 B	Reasoning specialised variant via SFT and RL; isolates the effect of math focused post training

Phase one performs contamination detection with calibrated thresholds. We fully index three reasoning relevant training sources, namely the Instella GSM8K synthetic dataset, the dm math corpus, and a curated subset of OpenMathInstruct 2, using a compact sentence encoder and FAISS. For the larger web corpora DCLM and FineWeb Edu we apply a content classifier and URL filter to retain mathematics and reasoning oriented passages before sampling, and we explicitly report estimated coverage rather than claim exhaustive search. Decision thresholds for cosine similarity and 13 token overlap are calibrated against a small held out paraphrase set constructed from benchmark items and unrelated passages, with reported precision and recall at the chosen operating point. Each benchmark item is then classified as contaminated, partially contaminated, or clean, with results from sampled sources interpreted as lower bounds.

Phase two establishes baseline accuracy. The two primary model variants generate chain of thought responses on every benchmark item under greedy decoding. Accuracy is reported disaggregated by contamination level, producing the headline accuracy gap between contaminated and clean items as the first diagnostic.

Phase three computes the Reliability Score. For 300 stratified problems we construct two semantically equivalent variants per problem, using numerical perturbation and irrelevant context injection, the two perturbation types with the strongest documented effects [2]. Let c_i be a binary indicator of whether variant i in a cluster is correct, computed after a benchmark specific canonicalisation step that normalises numerical formats and multiple choice outputs. Cluster accuracy is the mean of c_i . Cluster consistency is the fraction of variants whose correctness label matches that of the original problem. Reliability is the product of accuracy and consistency. This definition gives a uniformly incorrect cluster a reliability of zero, isolating the fragile pattern matching regime where the original is correct but variants are not.

Phase four performs lightweight training data attribution. The embedding index from phase one provides ranked nearest neighbours for every benchmark item, characterising whether the closest training passages are near duplicates, thematically related, or unrelated. As a focused validation step on AMD OLMo 1B, which fits in full precision on the available GPU, we compute TracIn checkpoint proximity scores for 100 selected items prefiltered through FAISS [4], to test whether embedding nearest neighbour attribution agrees with gradient based attribution on shared items. Full gradient attribution on the 3B variants is deferred to future work with larger compute. Phase five compares Instella 3B and Instella 3B Math to isolate the effect of math focused post training on reliability across subskills, following the metric robust framing of [5].

Reasoning Subskill	Benchmark	Approx. Items
Arithmetic word problems	GSM8K	1,319
Competition mathematics	MATH, levels 1 to 3	1,500
Logical deduction	LogiQA 2.0, subset	800
Compositional reasoning	CLUTRR, held out compositions	500

EXPECTED RESULTS

We expect three primary outcomes within the scoped compute budget. First, contamination rates against the fully indexed reasoning sources will be reported exhaustively, while contamination rates against sampled web corpora will be reported as conservative lower bounds with stated coverage. Second, the Reliability Score is expected to reveal a measurable gap between accuracy and reliability that varies by subskill, with contaminated items showing higher accuracy but lower consistency than clean items of comparable difficulty. Third, embedding based attribution profiles are expected to distinguish concentrated training data signatures, characteristic of memorisation, from diverse signatures, characteristic of generalisation, providing per subskill diagnostics. The validation experiment on the smaller open companion will indicate whether embedding nearest neighbour attribution is a sufficient proxy for gradient based attribution at this scale, informing whether the framework can later be scaled to richer compute. The unified deliverable is a focused diagnostic profile across four reasoning subskills, annotated with accuracy, consistency, reliability, and the most influential training data sources.

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